ICS 604
Applied Data Science

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Reminder:

- Homework Assignment 3
  - Due: **11:59 PM** on **Monay, April 6**
  - Submit only the Jupyter Notebook file (.ipynb) to Lamaku.
    - Make sure to **rename your notebook file** as instructed.
- Exercise 4 (chi-squared tests) is due today (Monday, March 30).
  - Upload completed notebook file to “Exercises” under Assignments in Lamaku.
- Project proposal peer feedback is due today.
  - Upload two PDF files to “Proposal Peer Feedback” under Assignments in Lamaku.

Announcements

- The notebook file for common hypothesis tests has been updated.
  - Old: ics604-17_hypothesis_testing_common_tests.ipynb
  - Updated: ics604-17-18_hypothesis_testing_common_tests.ipynb
  - The old file has been deleted. Please download the updated version.
- Correction
  - On page 17 of the Lecture 15 slides, the annotation for a Type I error should be corrected to FP (false positive).

Lecture 18

- Hypothesis Testing: Common Tests
  - z-test and t-tests
    - One sample z-test
    - One sample t-test
    - The independent samples t-test (Student's t-test)
    - The independent samples t-test (Welch's t-test)
- Correlations
  - Pearson's correlation coefficient (R)

z-test and t-test

- The z-test and t-test help us determine whether the differences between groups is statistically significant.
  - I.e., is the observed difference likely due to random variation (i.e., by chance)?
- Approach similar to the simulation-based hypothesis testing methods we've covered.
- Used for example to:
  - In a medical context, we might want to know if a new drug significantly increases or decreases blood pressure.
  - Compare the average transaction value between two companies.

One sample z-test

- Average expression level of gene X in normal corals is $\mu = 67.5$ with $\sigma = 9.5$.
- You measure the expression of gene X in 20 survivors of a heat wave and obtain the following values:

  50 60 60 64 66 66 67 69 70 74 76 76 77 79 79 79 81 82 82 89
- Biologically, we may be interested in evaluating whether the expression of gene X in survivors is different from that in normal corals.
  - Perhaps that is the reason those corals survived the heat wave?

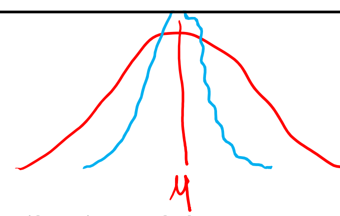
```
new_data = [50, 60, 60, 64, 66, 66, 67, 69, 70, 74, 76, 76, 77, 79, 79, 79, 81, 82, 82, 89]

np.mean(new_data)
```

72.3 — sample mean $\bar{X}$

One sample z-test (cont'd.)

- We are interested in whether there is a difference between the means.
  - I.e., the average gene expression in normal corals and the survivors.
- Test statistic: $\bar{X} - \mu$
  - If difference is close to zero, things are looking good for the null hypothesis.
  - If this quantity is too large or too small, then it's looking less likely that the null hypothesis is true.
- How far away from zero should it be for us to reject H_0?

z-score



- If the null hypothesis is true, then the sampling distribution of the mean can be written as follows:

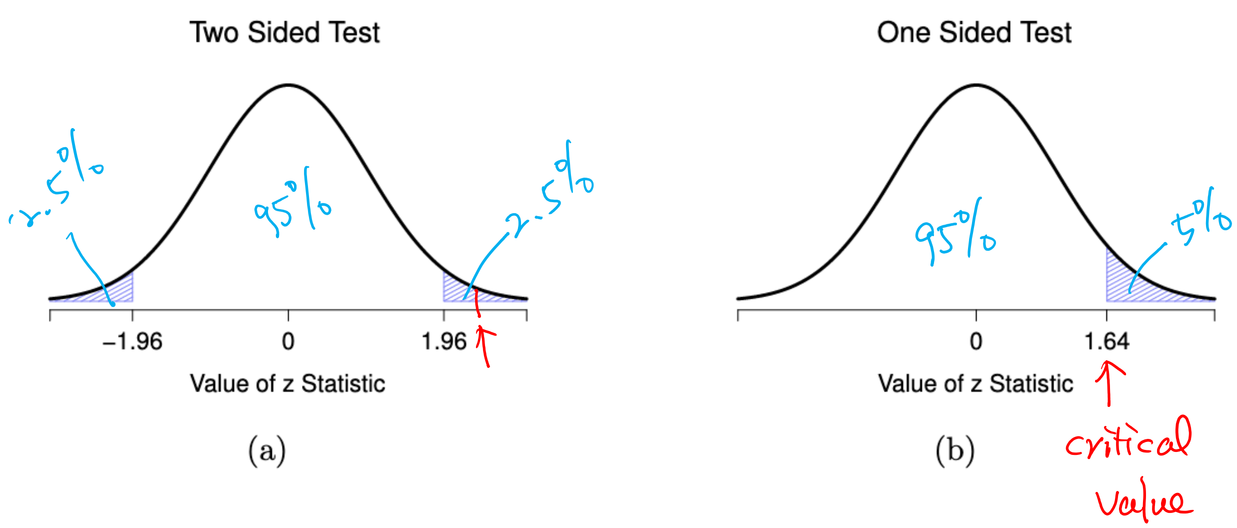
$$\bar{X} \sim \mathcal{N}(\mu, SE(\bar{X}))$$

where $SE(\bar{X}) = \sigma/\sqrt{N}$ (the standard error of $\bar{X}$). SEM

- What we can do is to convert the sample mean $\bar{X}$ into a *standard score*:

$$z_X = \frac{\bar{X} - \mu}{SE(\bar{X})} = \frac{\bar{X} - \mu}{\sigma/\sqrt{N}}$$

- This z-score is our test statistic.
  - The **z-scores** has a **standard normal distribution**: $z_X \sim \mathcal{N}(0,1)$




```python
sample_mean = np.mean(new_data)
sample_mean
```

```
72.3
```

```python
mu_null = 67.5
sd_true = 9.5
```

} pop.

```python
N = len(new_data)
sem = sd_true / np.sqrt(N)
z_score = (sample_mean - mu_null) / sem
z_score
```

```
2.259605535157681
```

observed stat.

```python
rv_norm = scipy.stats.norm(0, 1)
1 - rv_norm.cdf(z_score)
```

```
0.011922871882469877
```

```python
rv_norm.cdf(-z_score)
```

```
0.011922871882469896
```

```python
rv_norm.cdf(-z_score) + (1 - rv_norm.cdf(z_score))
```

```
0.023845743764939774
```

// p-value

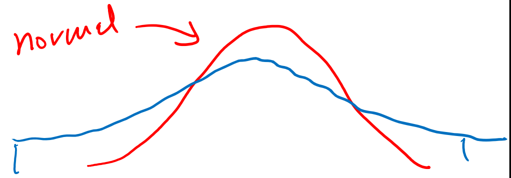
Conclusion of z-test for the example

- With an average expression of 72.3 of gene X in the sample of survivors, and assuming a true population standard deviation of 9.5, we can conclude that the survivors have significantly different statistics scores to the average of normal corals ($z = 2.26$, $N = 20$, $p < 0.05$).

Assumptions of the z-test

- Normality
  - The z-test assumes that the true population distribution is normal.
- Independence
  - The observations in the data are not correlated with each other.
- Known standard deviation
  - The true standard deviation of the population is known.

One sample t-test



- We don't know the population's standard deviation.
  - We need to adjust for the fact that we have some uncertainty about what the true population standard deviation is.
- If our null hypothesis is that the true mean is μ, but our sample has mean $\bar{X}$ and our estimate of the population standard deviation is $\hat{\sigma}$, then our t-statistic is:

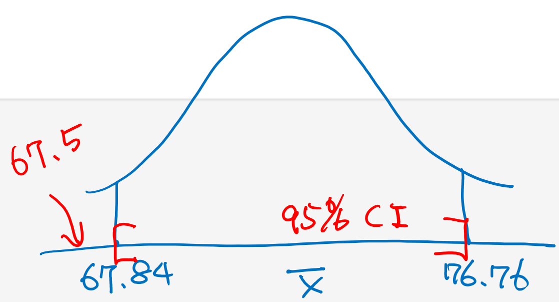
$$t = \frac{\bar{X} - \mu}{\hat{\sigma}/\sqrt{N}}$$

- The sampling distribution turns into a t-distribution with $N - 1$ degrees of freedom (df).

```
res = scipy.stats.ttest_1samp(new_data, popmean=67.5)
print("statistic:", round(res.statistic.item(), 3))
print("pvalue    :", round(res.pvalue.item(), 3))
print("df        :", round(res.df.item(), 1))
```

```
statistic: 2.255
pvalue    : 0.036
df        : 19
```

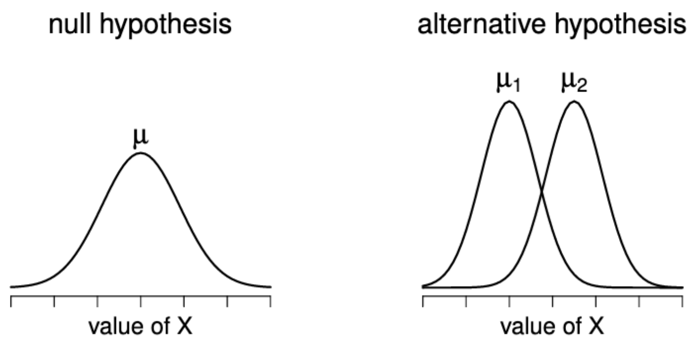
```
conf_level = 0.95
df = len(new_data) - 1
sample_mean = np.mean(new_data)
sem = scipy.stats.sem(new_data)

conf_interval = scipy.stats.t.interval(conf_level, df, sample_mean, sem)
tuple(round(x.item(), 2) for x in conf_interval)
```




```
(67.84, 76.76)
```

The Independent samples t-test (Student's t-test)

- Two independent sets of continuous values.
- Are the two independent samples drawn from populations with the same mean (null hypothesis) or different means (alternative hypothesis)?
  - H_0: $\mu_1 = \mu_2$
  - H_A: $\mu_1 \neq \mu_2$



Pooled estimate of the standard deviation

- We make the assumption that the two groups have the same population standard deviation.
  - $\sigma_1 = \sigma_2 = \sigma$
- Compute the *pooled standard deviation* simply as the weighted average across both samples:

$$\hat{\sigma} = \sqrt{\frac{(N_1 - 1)\hat{\sigma}_1^2 + (N_2 - 1)\hat{\sigma}_2^2}{N_1 + N_2 - 2}}$$

Test statistic

$$t = \frac{\bar{X}_1 - \bar{X}_2}{SE(\bar{X}_1 - \bar{X}_2)}$$

where

$$SE(\bar{X}_1 - \bar{X}_2) = \hat{\sigma}\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}$$

- The sampling distribution of this t-statistic is a t-distribution with $N - 2$ degrees of freedom.

$= (N_1 - 1) + (N_2 - 1) = \underbrace{N_1 + N_2}_{N} - 2$

```
np.random.seed(142)
data_1 = np.random.normal(1, 3, 20)
data_2 = np.random.normal(1, 3, 50)
```

$\} \; 70$

```
# Student's t-test
res = scipy.stats.ttest_ind(data_1, data_2)
print("statistic:", round(res.statistic.item(), 4))
print("pvalue    :", round(res.pvalue.item(), 4))
print("df        :", round(res.df.item(), 1))
```

```
statistic: 1.2033
pvalue    : 0.233
df        : 68.0
```

$= 70 - 2$

Assumptions of the Student's t-test

- Normality
  - Each group's values are approximately normally distributed
- Independence
  - Observations are independent within and between groups.
- Homogeneity of variance (also called "homoscedasticity")
  - Population standard deviations are equal: $\sigma_1 = \sigma_2$

The independent samples t-test (Welch's t-test)

- Two independent sets of continuous values.
  - Are the two independent samples drawn from populations with the same mean (null hypothesis) or different means (alternative hypothesis)?
- Both groups may not have the same standard deviation.
  - If two samples don't have the same means, why should we expect them to have the same standard deviation?
- Does not assume homogeneity of variance.
  - This leaves only the assumption of normality, and the assumption of independence.

Test statistic

$$t = \frac{\bar{X}_1 - \bar{X}_2}{SE(\bar{X}_1 - \bar{X}_2)}$$

where

$$SE(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\hat{\sigma}_1^2}{N_1} + \frac{\hat{\sigma}_2^2}{N_2}}$$

- The standard error reflects the combined variability of both samples.
  - Each sample contributes its own uncertainty, based on its variance and size.

```
data_1 = np.random.normal(4, 0.8, 20)
data_2 = np.random.normal(1, 1.2, 50)
```

```
# Welch's t-test
res = scipy.stats.ttest_ind(data_1, data_2, equal_var=False)
print("statistic:", round(res.statistic.item(), 4))
print("pvalue    :", round(res.pvalue.item(), 4))
print("df        :", round(res.df.item(), 1))
```

```
statistic: 9.1952
pvalue    : 0.0
df        : 33.3
```

Correlations

Correlations

- Given n observations from two random variables X and Y.
- We say that X and Y are *correlated* if changes in one are related to changes in the other.
  - The *correlation coefficient* between X and Y is a numerical value that quantifies the strength and direction of the relationship between the two variables.

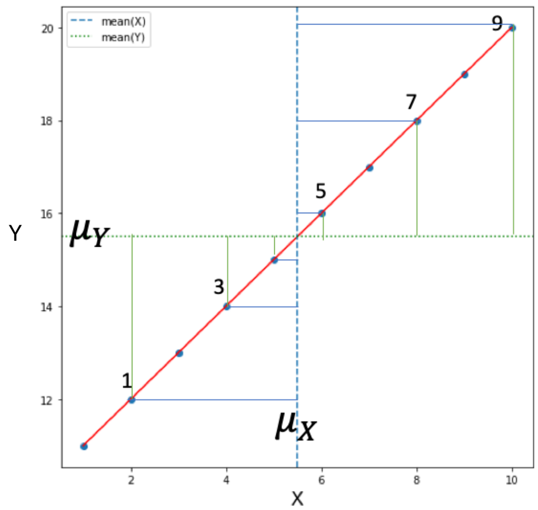
Correlations: Examples

- Weight is positively correlated with height (relatively strong correlation).
- The number of tourist in Waikiki is positively associated with total sales at ABC stores (reasonably strong correlation).
- IQ is negatively correlated with the time required to solve simple logic problems (moderate correlation).
- Daily cigarette consumption is negatively correlated with years lived (reasonably strong correlation).
- The number of video games played is weakly positively correlated with violent behavior scores (very weak and not easily reproducible).

Linear correlations

- Linear correlation ranges from –1 (*perfect anti-correlation*) to 1 (*perfect positive correlation*), with 0 indicating no linear correlation.
- One of the most widely used measures of linear correlation is **Pearson's Correlation Coefficient (R)**.
- We will also introduce **Spearman's Rank Correlation (ρ)**, which measures the strength and direction of a monotonic relationship.

Positive correlation and deviation from the means



	X	Y
0	1	11
1	2	12
2	3	13
3	4	14
4	5	15
5	6	16
6	7	17
7	8	18
8	9	19
9	10	20

$Y = X + 10$

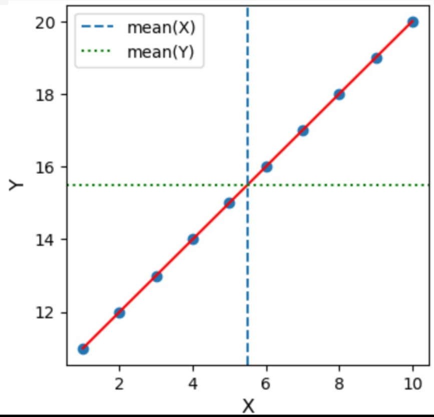
```python
# Example of a deterministic relationship: Y = X + 10

X = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Y = [11, 12, 13, 14, 15, 16, 17, 18, 19, 20]

data = pd.DataFrame({"X": X, "Y": Y})
data
```

	X	Y
0	1	11
1	2	12
2	3	13
3	4	14
4	5	15
5	6	16
6	7	17
7	8	18
8	9	19
9	10	20

```python
print(data["X"].mean(), data["Y"].mean())
```

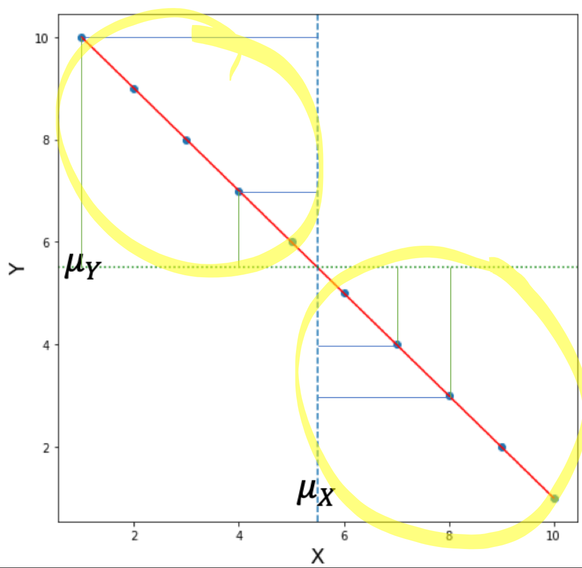
```
5.5 15.5
```

```python
plt.figure(figsize=(4, 4))

plt.scatter(X, Y)
plt.plot(X, Y, color='r')

plt.xlabel("X", fontsize=12)
plt.ylabel("Y", fontsize=12)
plt.axvline(data["X"].mean(), linestyle='--', label="mean(X)")
plt.axhline(data["Y"].mean(), linestyle=':', label="mean(Y)", color='green')
plt.legend();
```



In this example, both X_i and Y_i are increasing – X and Y are positively correlated.

Negative correlation and deviation from the means



	X	Y
0	1	10
1	2	9
2	3	8
3	4	7
4	5	6
5	6	5
6	7	4
7	8	3
8	9	2
9	10	1

$Y = -X + 11$

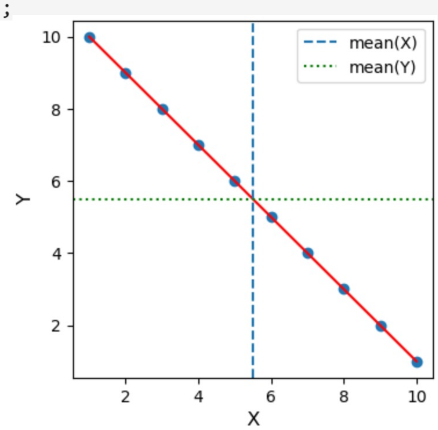
```python
X = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Y = [10, 9, 8, 7, 6, 5, 4, 3, 2, 1]

data = pd.DataFrame({"X": X, "Y": Y})
data
```

	X	Y
0	1	10
1	2	9
2	3	8
3	4	7
4	5	6
5	6	5
6	7	4
7	8	3
8	9	2
9	10	1

```python
print(data["X"].mean(), data["Y"].mean())
```

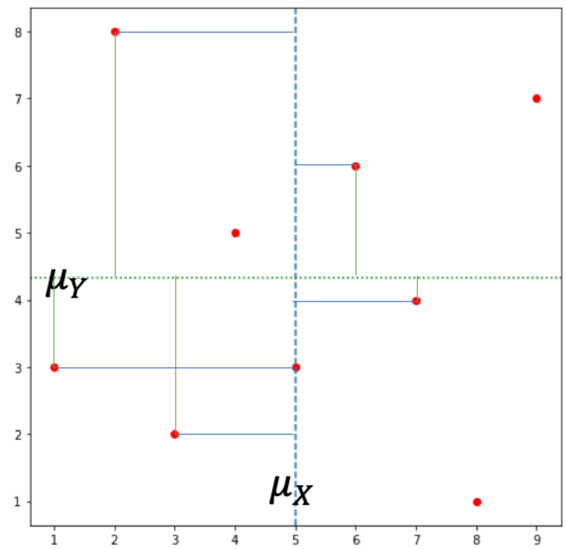
```
5.5 5.5
```

```python
plt.figure(figsize=(4, 4))

plt.scatter(X, Y)
plt.plot(X, Y, color='r')

plt.xlabel("X", fontsize=12)
plt.ylabel("Y", fontsize=12)
plt.axvline(data["X"].mean(), linestyle='--', label="mean(X)")
plt.axhline(data["Y"].mean(), linestyle=':', label="mean(Y)", color='green')
plt.legend();
```



In this example, Y_i is decreasing when X_i is increasing – X and Y are negatively correlated.

Lack of correlation and deviation from the means



	X	Y
0	1	3
1	2	8
2	3	2
3	4	5
4	5	3
5	6	6
6	7	4
7	8	1
8	9	7

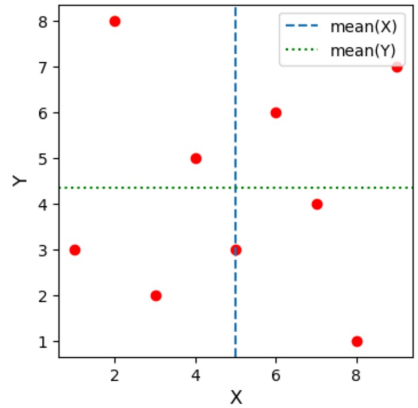
```
X = [1, 2, 3, 4, 5, 6, 7, 8, 9]
Y = [3, 8, 2, 5, 3, 6, 4, 1, 7]

data = pd.DataFrame({"X": X, "Y": Y})
data
```

	X	Y
0	1	3
1	2	8
2	3	2
3	4	5
4	5	3
5	6	6
6	7	4
7	8	1
8	9	7

```
print(data["X"].mean(), data["Y"].mean())
```

```
5.0 4.333333333333333
```

```python
plt.figure(figsize=(4, 4))

plt.scatter(X, Y, color='r')

plt.xlabel("X", fontsize=12)
plt.ylabel("Y", fontsize=12)
plt.axvline(data["X"].mean(), linestyle='--', label="mean(X)")
plt.axhline(data["Y"].mean(), linestyle=':', label="mean(Y)", color='green')
plt.legend();
```



Covariance

(X_i, Y_i)

- Measure of the tendency of two variables to vary together.
- Product of deviations from the means: $(X_i - \bar{X})(Y_i - \bar{Y})$
  - Positive when the deviations have the same sign.
  - Negative when the deviations have the opposite sign.
- Adding up the products of deviations gives a measure of the tendency to vary together.
  - Covariance is the average of these products:

$$Cov(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})$$

$$\begin{array}{c|cc} & X & Y \\ \hline X & cov(X,X) & cov(X,Y) \\ Y & cov(Y,X) & cov(Y,Y) \end{array}$$

- Covariance is the dot product of the deviations, divided by their length.
  - Maximized if the two vectors are identical, 0 if they are orthogonal, and negative if they point in opposite directions.

Pearson's correlation coefficient

- Covariance is useful in some computations, but it is seldom reported as a summary statistic.
  - An issue: unit of the covariance is the product of the units of X and Y (e.g., pounds(weight)-inches(height)).
  - A solution: compute the product of standard scores (deviations divided by the standard deviation).

$$r_i = \frac{(X_i - \bar{X})}{\hat{\sigma}_X} \cdot \frac{(Y_i - \bar{Y})}{\hat{\sigma}_Y}$$

- Pearson's Correlation Coefficient: the mean of the products above

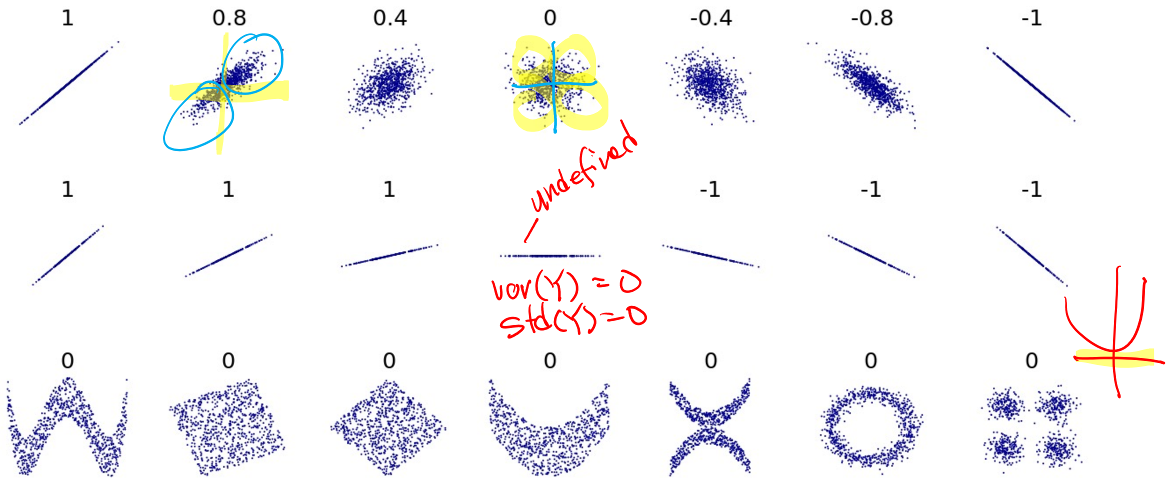
$$R = \frac{1}{n-1}\sum_{i=1}^{n} r_i = \frac{Cov(X,Y)}{\hat{\sigma}_X \cdot \hat{\sigma}_Y} = \frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

Pearson's correlation coefficient (cont'd.)

$$R = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}}$$

- If the correlation is positive: when X's distance from its mean increases, Y's distance from its mean increases as well.
- If the correlation is negative: when X's distance from its mean increases, Y's distance from its mean decreases (increases in the negative direction) or vice-versa.
- In the absence of correlation, the contributions from X's and Y's cancel out. Therefore, the correlation tends toward 0.

Pearson's correlation coefficient: Examples



https://en.wikipedia.org/wiki/Correlation